Sentiment Analysis of LLM User Responses by Conversation Length and User Query Type

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Abstract

Large Language Models (LLMs) are becoming increasingly integrated into everyday life and professional workflows. The WildChat dataset, which contains approximately one million conversations between users and ChatGPT, provides a rich resource for analyzing how users interact with LLMs at scale. This report uses the WildChat dataset to examine user satisfaction across different types of LLM conversations, using the VADER compound sentiment score as a proxy for satisfaction with the model's responses. We address two research questions: (1) how many conversational turns users typically engage in with ChatGPT, and whether user satisfaction differs between shorter and longer conversations; and (2) which categories of user queries are associated with higher or lower post-response satisfaction.

Code — https://github.iu.edu/noldfath/26SP-ILS-Z639-SMM/blob/main/Nick-Oldfather-assignment1.ipynb

Introduction

In recent years, large language models (LLMs) have become increasingly commonplace and are often used as an advanced alternative to traditional search engines. Owing to their ability to interpret fine-grained details and engage in human-like conversation, LLMs have gained widespread adoption in domains such as mathematics, programming, and general problem-solving. Despite their strong performance, LLMs are not without limitations: they may produce hallucinated or incorrect answers, or respond in formats that do not align with user expectations.

Such failures can be frustrating for users, motivating a closer examination of where these pain points occur and whether systematic patterns in user dissatisfaction can be identified. This report investigates user satisfaction in interactions with ChatGPT by addressing two research questions: (1) how many conversational turns users typically engage in per session, and whether user satisfaction differs between shorter and longer conversations; and (2) whether certain types of user questions are associated with higher or lower satisfaction with the LLM's responses.

To address these questions, we analyze the WildChat dataset, which contains approximately one million real-world conversations between users and ChatGPT and spans over 2.5 million user turns. The dataset spans a diverse range of topics and conversation lengths, making it well-suited for large-scale analysis of naturalistic LLM interactions [4]. In addition, WildChat provides structured information about the number of conversational turns within each session, enabling analysis of interaction length.

Because WildChat doesn't include explicit labels for question types or user reactions, this study relies on approximate methods. Natural language processing techniques, including text vectorization and clustering, are used to group semantically similar user queries. User satisfaction is approximated using VADER's compound sentiment score [2], applied to user turns following LLM responses.

Related Work

Prior work by Zhang et al analyzes sentiment dynamics in the WildChat dataset, focusing on how user sentiment evolves as conversations progress. Their analysis emphasizes changes in sentiment over time, rather than comparing aggregate satisfaction between shorter and longer conversations, which is the focus of this report.

They found that users tend to exhibit increased frustration in conversations lasting five or more turns. Following their findings, we adopt five turns as the threshold distinguishing short and long conversations. Their study relies on sentiment labels generated by both trained professionals and LLMs, whereas this report employs automated sentiment analysis and unsupervised clustering methods due to constraints on time and resources. While these methods are likely less robust than human annotation, they nonetheless provide a scalable approximation for exploratory analysis.

Additionally, prior work observes that conversations often span multiple topics within a single session [3]. Consequently, for research question two, this study clusters each turn by the user independently. For example, a user may ask a programming question followed immediately by a historical question in subsequent turns of the same conversation; treating each turn separately allows these queries to be assigned to different semantic clusters.

Dataset and Data Processing

The WildChat dataset was collected between April 2023 and April 2024 using two models: ChatGPT-3.5-Turbo and GPT-

4. Users voluntarily consented to have their LLM interactions recorded in exchange for free access to the models.

The WildChat dataset has several limitations. Both chatbot services were hosted on Hugging Face Spaces, which likely resulted in a user base skewed toward programmers and technically inclined users. As a result, the queries may be less diverse and less representative of the general public. Additionally, users were not required to provide identifying information, which contributed to a higher prevalence of potentially toxic queries. However, conversations flagged as toxic or containing private data were excluded from the released dataset [4]. Overall, the dataset's collection method and participant population introduce inherent biases that may limit generalizability.

The dataset contains fourteen fields, but only four were used in this analysis: `conversation_hash`, `conversation`, `turn`, and `language`. The `conversation_hash` serves as a unique identifier for a conversation's text; identical conversations may share the same hash if they occurred in separate sessions. The `conversation` field contains the full text of a session, including both user prompts and assistant responses, and may span multiple turns. The `turn` field indicates the total number of conversational turns within a session; for example, a two-turn conversation consists of two user prompts and two corresponding LLM responses. The `language` field specifies the language of the conversation.

Due to the dataset's large size (approximately 3.4 GB), only the first 50,000 rows were included in the analysis. All selected conversations were recorded in April 2023, which may result in overrepresentation of topics or events relevant to that time period.

Non-English conversations were excluded, as the VADER sentiment analyzer performs best on English text [2], and multilingual topic modeling was beyond the scope of this study. Additionally, duplicate conversations sharing the same `conversation_hash` were removed, ensuring that identical interactions were only counted once. After filtering, 23,698 unique conversations remained for analysis.

Research Question 1 (Method, Findings)

The first research question this report addresses is how many turns users typically spend conversing with the LLM per session, and whether user satisfaction differs between shorter and longer conversations. Figures 1 and 2 show that the distribution of conversation lengths is heavily right-skewed, with the majority of conversations consisting of only a few turns. Over 50% of conversations have just one turn. Summary statistics are as follows: lower whisker = 1, Q1 = 1, median = 1, Q3 = 3, and upper whisker = 6.

Single-turn conversations were excluded because the user's initial query does not reflect sentiment toward the LLM's response. Similarly, the first user turn was removed from multi-turn conversations, and assistant responses were excluded, leaving only user turns after the first for analysis.

Conversations with fewer than four turns were classified as short, and those with four or more as long. Of the 11,220 eligible conversations, 68.7% were short, and 31.3% were

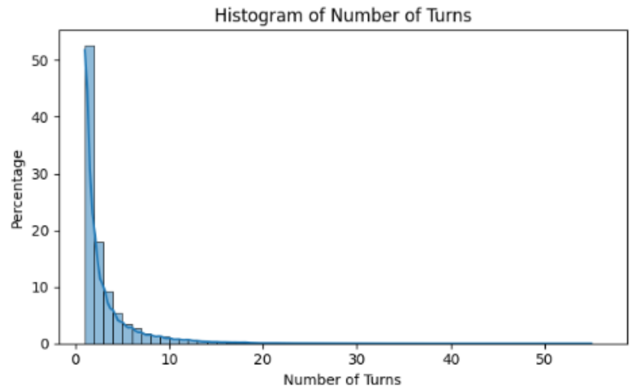

Figure 1: Histogram of Number of Turns for Conversations

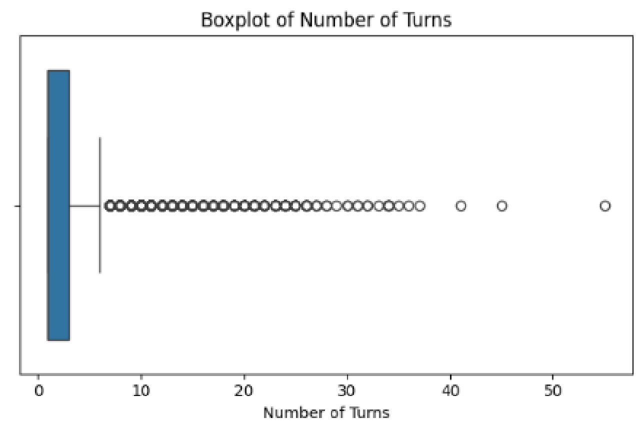

Figure 2: Boxplot of Number of Turns for Conversations

long. Sentiment for each user turn was scored using VADER (Valence Aware Dictionary and sEntiment Reasoner) and then averaged across conversations. VADER is a rule-based system (not machine learning) that scores words from -4 to +4 and normalizes text-level scores to -1 to 1, with $<$ -0.05 negative, -0.05 to 0.05 neutral, and $>$ 0.05 positive. It accounts for punctuation, capitalization, degree modifiers (e.g., "very good" vs. "good"), and performs well on short texts without requiring training, making it suitable for WildChat messages [2].

Figure 3 shows that short conversations have a median score of 0, while long conversations have a median of 0.1. A Mann-Whitney U test confirms this difference is significant ($p < 0.001$). Short conversations exhibit greater variability (SD = 0.376 vs. 0.235), suggesting they are more polarizing: some users quickly end conversations out of frustration, while others swiftly find what they need. Short conversations also show higher mean (0.153 vs. 0.134), third quartile (0.361 vs. 0.254), and upper whisker (0.903 vs. 0.636) scores, but a lower lower whisker (-0.542 vs. -0.382).

Overall, longer conversations tend to have higher median sentiment and less variance. One possible explanation is that users express sentiment more consistently in shorter sessions. While in longer sessions, sentiment may appear only

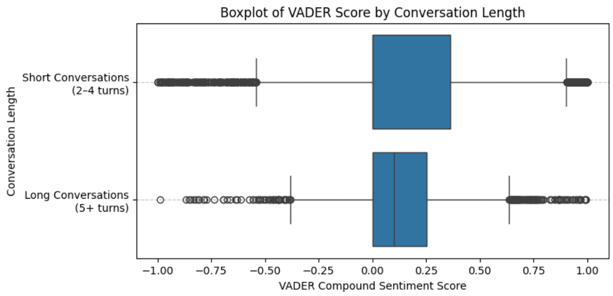

Figure 3: Boxplot of VADER Scores by Conversation Length

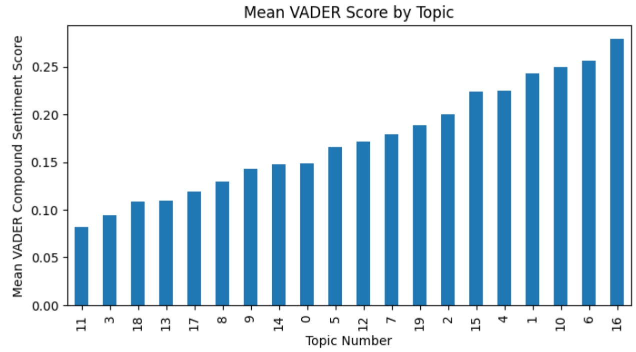

Figure 4: Mean VADER Score by Topic

in a few turns, with neutral language dominating the rest. This could be due to users being less fatigued in shorter conversations and being willing to use expressive language more frequently. However, this explanation is speculative and warrants further study.

Research Question 2 (Method, Findings)

The second research question this report seeks to answer is which categories of user questions are associated with higher or lower levels of post-response satisfaction.

Previous work with the WildChat dataset by Zhang et al. found that multi-turn conversations often span a variety of topics, meaning each turn may represent a different type of question. They also found that the subsequent user turn can reflect sentiment toward the LLM's response to the previous turn [3].

VADER compound sentiment scores were used to estimate user sentiment [2]. Scores were backshifted by one turn within the same conversation, so the sentiment of turn 2 was paired with turn 1's question, reflecting the user's reaction to the LLM's response. Turn 1 data for multi-turn conversations was reintroduced for analysis, and the final turn was dropped, as there was no subsequent turn to pair sentiment with. One-turn conversations were still excluded, as no subsequent sentiment could be derived.

Because questions were not pre-labeled by type, natural language processing was used to group them. Each turn was vectorized, ignoring common English words that do not indicate question type. After tuning, vectors were clustered using K-Means into 20 clusters, selected via the Elbow method and Silhouette Score [1]. The five most frequent unique words for each cluster, along with the average VADER scores, are shown in Table 2 and Figure 4. Overall, VADER scores ranged from 0.082 to 0.280, indicating users generally have neutral to slightly positive sentiment towards LLM responses, regardless of user question type.

Table 1: Average VADER Scores by Topic - Topic Subtypes Based on Top 10 Words

Topic Number	Topic Subtype	Average VADER Score
0	Coding/Technical	0.148
1	Other	0.243
2	Other	0.200
3	Coding/Technical	0.094
4	Social Studies/Science	0.224
5	Writing/Conversational	0.166
6	Coding/Technical	0.256
7	Writing/Conversational	0.179
8	Other	0.130
9	Writing/Conversational	0.143
10	Writing/Conversational	0.249
11	Coding/Technical	0.082
12	Coding/Technical	0.171
13	Language Translation	0.109
14	Coding/Technical	0.148
15	Social Studies/Science	0.224
16	Social Studies/Science	0.280
17	Other	0.119
18	Coding/Technical	0.108
19	Writing/Conversational	0.189

Table 2: 5 Most Frequent Words by Topic - Check GitHub for Additional Words

Topic Number	Most Common Words
0	{'code', 'write', 'python', 'java', 'run'}
1	{'best', 'write', 'try', 'place', 'list'}
2	{'time', 'people', 'game', 'think', 'work'}
3	{'different', 'class', 'try', 'wrong', 'div'}
4	{'example', 'ni', 'hermanubis', 'god', 'names'}
5	{'generate', 'women', 'chatgpt', 'character', 'prompt'}
6	{'file', 'args', 'plan', 'memory', 'py'}
7	{'hi', 'chat', 'app', 'memory', 'chatgpt'}
8	{'list', 'risks', 'provide', 'team', '10'}
9	{'write', 'story', 'write story', 'article', 'funny'}
10	{'good', 'morning', 'think', 'bad', 'english'}
11	{'does', 'mean', 'work', 'exist', 'code'}
12	{'image', 'new', 'int', 'string', 'public', 'np'}
13	{'translate', 'english', 'chinese', 'japanese', 'text'}
14	{'data', 'error', 'self', 'df', 'based'}
15	{'book', 'win', 'match', 'vs', 'title'}
16	{'model', 'population', 'density', 'diffusion', 'growth'}
17	{'hello', '10', 'know', 'need', 'error'}
18	{'explain', 'simple', 'language', 'terms', 'example'}
19	{'create', 'answer', 'question', 'correct', 'report'}

The lowest-sentiment group consists of language translation questions (Topic 13), with an average score of 0.109. This aligns with expectations, as learning a new language and handling mistranslations can be frustrating.

Next are the coding and technical questions (Topics 0, 3, 6, 11, 12, 14, 18), with an average score of 0.144. Removing Topic 6 (Python coding questions), which had relatively higher sentiment, lowers the average to 0.125. This is consistent with the fact that Python is generally considered user-friendly; its less challenging nature may lead to slightly higher sentiment, whereas more technical tasks can increase frustration.

Following these are writing and conversational tasks (Topics 5, 7, 9, 10, 19), with an average score of 0.185.

The highest-sentiment group consists of social studies and science questions (Topics 4, 15, 16), with an average score of 0.243.

The remaining topics (1, 2, 8, 17) cover a variety of subjects and have an average score of 0.173.

Overall, the results suggest that more technical and challenging topics tend to be associated with lower user sentiment. Questions involving learning a new language, coding, or math are naturally more demanding and may lead to greater frustration. In contrast, less technical or more enjoyable tasks, such as story writing, exploring mythology, or learning about population trends, tend to elicit higher sentiment scores. In general, as question complexity increases, user sentiment decreases.

Statistical comparisons of VADER scores between topics are shown in Figure 5, with p-values for Mann-Whitney U Tests adjusted for multiple comparisons.

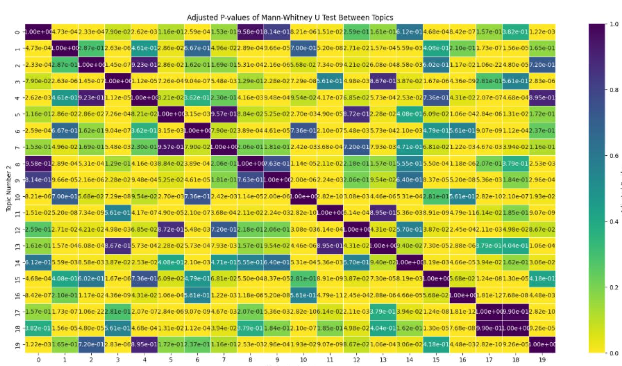

Figure 5: Adjusted Significance Test for Topic Mean VADER Scores

Conclusion

This report uses the WildChat dataset and VADER scores to examine the distribution of turns users spend conversing with LLMs, whether conversation length affects user sentiment, and which types of questions are associated with higher or lower sentiment.

Conversations were generally short, typically lasting one to three turns. Longer conversations had slightly higher median sentiment scores, while shorter conversations were more polarizing, exhibiting both more positive and more negative sentiment. This may be due to users using less expressive language in longer conversations, though further research would be needed to confirm.

Overall, users were neutral to slightly positive across most topics of conversation. However, sentiment tended to be lower for questions involving more technical or complex topics.

It is important to reiterate that the WildChat dataset is biased based on how the data was collected and may not generalize to the broader population. Additionally, VADER scores provide only a rough estimate of user sentiment toward LLM responses. A more precise measure would require alternative methods, such as asking users to rate their reactions to individual responses or to the session as a whole.

References

[1] *Clustering Text Documents using K-Means*. scikit-learn.org.
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[2] Hutto, C.J., & Gilbert, Eric. (2014, May 16). *VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text*. ICWSM.
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[4] Zhao Wenting, Ren Xiang, Hessel Jack, Cardie Claire, Choi Yejin, & Deng, Yuntian. (2024, May 2). *WildChat: 1M ChatGPT Interaction Logs in the Wild*. arXiv.org.
https://arxiv.org/abs/2405.01470

Appendix

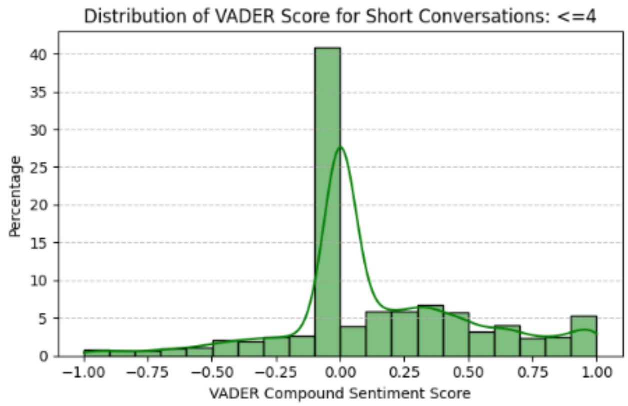

Figure A1: Distribution of VADER Scores for Short Conversations

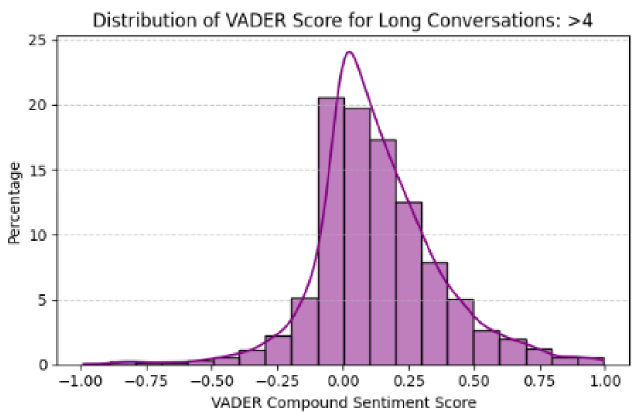

Figure A2: Distribution of VADER Scores for Long Conversations

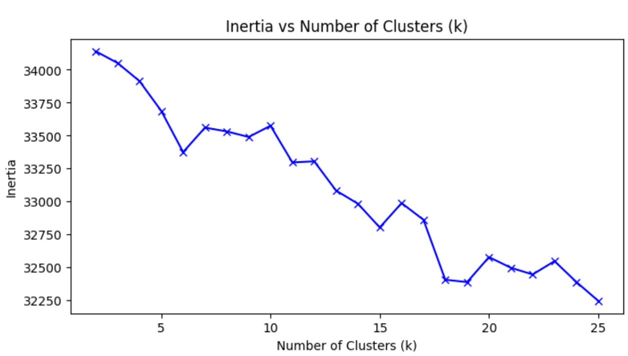

Figure A3: Chart of Inertia vs Number of Clusters

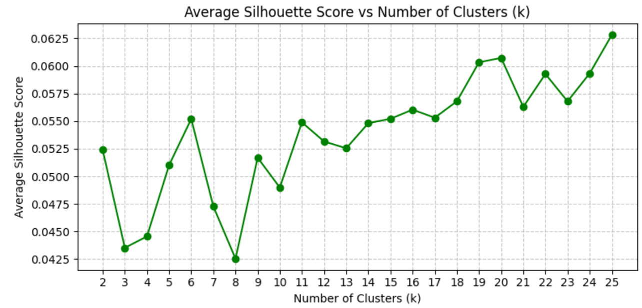

Figure A4: Chart of Average Silhouette Score vs Number of Clusters

Table A1: Number of User Queries Assigned to Each Topic

Topic Number	Number of Turns Per Topic
0	870
1	341
2	2512
3	725
4	609
5	1497
6	588
7	421
8	659
9	2178
10	314
11	562
12	835
13	214
14	712
15	469
16	402
17	23832
18	478
19	794